

Geometria Simplética 2022, Lista 1

professor: Henrique Bursztyn

deadline: 30/08

Problem 1: Let V be a symplectic vector space ($\dim(V) = 2n$), and $\Omega \in \wedge^2 V^*$ be a skew-symmetric bilinear form. Show that Ω is nondegenerate iff $\Omega^n \neq 0$.

Problem 2: Let (V, Ω) be a symplectic vector space, and let $W \subseteq V$ be any linear subspace.

- a) Show that $V_W := \frac{W}{W \cap W^\Omega}$ inherits a natural symplectic structure Ω_W uniquely determined by the condition $\pi^* \Omega_W = \Omega|_W$ (here $\pi : W \rightarrow W/(W \cap W^\Omega)$ is the quotient projection).
(The space (V_W, Ω_W) is called the “reduced space”.)
- b) Suppose that W is *coisotropic*, and let $L \subset V$ be lagrangian. Show that the image of $L \cap W$ via $\pi : W \rightarrow V_W$ is lagrangian in the reduced space.

Problem 3: We saw in class that any symplectomorphism $T : V_1 \rightarrow V_2$ defines a lagrangian subspace by its graph: $\Gamma_T := \{(Tu, u), u \in V_1\} \subset V_2 \oplus \bar{V}_1$. So we think of lagrangian subspaces of $V_2 \oplus \bar{V}_1$ as generalizations of symplectomorphisms. We now see how to generalize their composition.

Consider symplectic vector spaces V_1, V_2, V_3 , and $E = V_3 \oplus \bar{V}_2 \oplus V_2 \oplus \bar{V}_1$.

- a) Show that $\Delta := \{(v_3, v_2, v_2, v_1) \in E\}$ is coisotropic in E and its reduction E_Δ can be identified with $V_3 \oplus \bar{V}_1$.
- b) Given lagrangian subspaces $L_1 \subset V_2 \oplus \bar{V}_1$ and $L_2 \subset V_3 \oplus \bar{V}_2$, define the *composition* of L_2 and L_1 by

$$L_2 \circ L_1 := \{(v_3, v_1) \mid \exists v_2 \in V_2 \text{ s.t. } (v_3, v_2) \in L_2, (v_2, v_1) \in L_1\}.$$

Show that $L_2 \circ L_1$ is a lagrangian subspace of $V_3 \oplus \bar{V}_1$. (Hint: show that the composition can be identified with the reduction of $L_2 \times L_1 \subset E$ with respect to Δ).

- c) Let $T_1 : V_1 \rightarrow V_2$ and $T_2 : V_2 \rightarrow V_3$ be symplectomorphisms. Show that $\Gamma_{T_2 \circ T_1} = \Gamma_{T_2} \circ \Gamma_{T_1}$.

Problem 4: Let (V, J) be a complex vector space, let Ω be a symplectic structure on V . Show that J and Ω are compatible iff there exists a hermitian inner product $h : V \times V \rightarrow \mathbb{C}$ such that Ω is its imaginary part. Show that any (complex) orthonormal basis of (V, h) can be extended to a symplectic bases of (V, Ω) .

Problem 5: Consider the symplectic vector space $(\mathbb{R}^{2n}, \Omega_0)$, where $\Omega_0(u, v) = -u^t J_0 v$ (same notation as in class). Check that its group of linear symplectomorphisms is given by $Sp(2n) = \{A \in GL(2n) \mid A^t J_0 A = J_0\}$. Show that $Sp(2n)$ is a smooth submanifold of $GL(2n)$ and that its tangent space at the identity $I \in GL(2n)$ is given by $T_I Sp(2n) = \{A : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n} \mid A^t J_0 + J_0 A = 0\}$. Conclude that $Sp(2n)$ has dimension $2n^2 + n$. Verify also that $Sp(2n)$ is not compact.

Problem 6: Consider the standard compatible triple (Ω_0, J_0, g_0) on $\mathbb{R}^{2n}$ (as in class). Let $O(2n)$ be the linear orthogonal group of $\mathbb{R}^{2n}$ (i.e, linear transformations preserving the canonical inner product g_0), and let $Sp(2n)$ be the symplectic linear group. Through the identification $\mathbb{R}^{2n} \cong \mathbb{C}^n$ (as complex vector spaces), we may see $GL(n, \mathbb{C})$ (the group of linear automorphisms of $\mathbb{C}^n$) as a subgroup of $GL(2n, \mathbb{R})$: a complex matrix $A + iB$ is identified with the real $2n \times 2n$ matrix

$$\begin{pmatrix} A & -B \\ B & A \end{pmatrix}.$$

Let now $U(n) \subset GL(n, \mathbb{C})$ be the group of linear transformation preserving the natural hermitian inner product of $\mathbb{C}^n$. Show that the intersection of any two of the groups

$$Sp(2n), O(2n), GL(n, \mathbb{C}) \subset GL(2n, \mathbb{R})$$

is $U(n)$.

Problem 7: Let (V, Ω) be a symplectic vector space, let $W \subseteq V$. Let J be a Ω -compatible complex structure, and g the corresponding inner product. Verify that $J(W^\Omega) = W^{\perp_g}$. (a) Use this fact to show that any coisotropic subspace of V has an isotropic complement. In particular, any lagrangian subspace $L \subset V$ has a lagrangian complement L' , $V = L \oplus L'$. (b) Show that there is a natural identification $L' \cong L^*$, that induces a symplectomorphism $V \cong L \oplus L^*$ (where $L \oplus L^*$ has the natural symplectic structure $((l, \alpha), (l', \alpha')) \mapsto \alpha(l') - \alpha'(l)$).

Bonus problem: Prove the following generalizations of problem 7 about lagrangian complements:

- (1) Let $W_1, \dots, W_k$ be lagrangian subspaces of V . Show that there is a lagrangian subspace $L \subset V$ satisfying $L \cap W_j = \{0\}$ for all j . [*Hint: problem 2 may help...*]
- (2) Let $E \subseteq V^{2n}$ be an arbitrary subspace of dimension n . Show that there is a *lagrangian* subspace L such that $E \oplus L = V$.

Geometria Simplética 2024, Lista 2

Professor: H. Bursztyn

Entrega dia 10/09

Problem 1: Verify (and justify) whether or not the following manifolds admit a symplectic structure: $S^1 \times S^3$, $\mathbb{R} \times S^3$, $\mathbb{R}^3 \times S^3$, $\mathbb{T}^3 \times S^3$.

Problem 2: Show that the tautological 1-form $\alpha \in \Omega^1(T^*Q)$ is *uniquely characterized* by the following property: for any 1-form $\mu \in \Omega^1(Q)$,

$$\mu^* \alpha = \mu,$$

where on the left-hand side we view μ as a map $\mu : Q \rightarrow T^*Q$.

Problem 3: We will characterize symplectomorphisms $T^*Q \rightarrow T^*Q$ which are cotangent lifts of diffeomorphisms $\phi : Q \rightarrow Q$. Let α be the tautological 1-form on $M = T^*Q$ and $\omega = -d\alpha$. We saw in class that cotangent lifts preserve α . We will show the converse of this fact.

Let $F : M \rightarrow M$ be a symplectomorphism such that $F^* \alpha = \alpha$.

- (a) Let $v \in \mathfrak{X}(M)$ be the unique vector field such that $i_v \omega = -\alpha$; note that, locally, it is given by $\sum_i \xi_i \frac{\partial}{\partial \xi_i}$ (v is known as the *Euler vector field*). Show that $F_* v = v$.
- (b) Let φ_t^v denote the flow of v . Show that $\varphi_t^v \circ F = F \circ \varphi_t^v$. Check that, in coordinates, $\varphi_t^v(x, \xi) = (x, e^t \xi)$, $-\infty < t < \infty$.
- (c) Verify that, for $p \in T_x^*Q$, $F(\lambda p) = \lambda F(p)$, $\forall \lambda \in \mathbb{R}$. Conclude that there exists $\phi : Q \rightarrow Q$ such that $\phi \circ \pi = \pi \circ F$ (here $\pi : T^*Q \rightarrow Q$ is the projection). Finally, show that $F = \widehat{\phi}$ (the cotangent lift of ϕ).

Problem 4: Let $\alpha \in \Omega^1(T^*Q)$ be the tautological 1-form. We will now see examples of symplectomorphisms of T^*Q which are not cotangent lifts. Let $A \in \Omega^1(Q)$ and consider the associated “fiber-translation” map $\varphi_A : T^*Q \rightarrow T^*Q$, $(x, \xi) \mapsto (x, \xi + A_x)$.

- (a) Show that

$$\varphi_A^* \alpha - \alpha = \pi^* A,$$

where $\pi : T^*Q \rightarrow Q$ is the projection. It follows that φ_A is a symplectomorphism iff A is *closed*.

- (b) Consider functions that are constant along the fibers of T^*Q (i.e., of the form $H = \pi^* f$, for $f \in C^\infty(Q)$). Describe their hamiltonian vector fields in local cotangent coordinates, as well as their flows.

Problem 5: Let $\omega = -d\alpha$ be the canonical symplectic form on T^*Q . Prove that, if $B \in \Omega^2(Q)$ is closed, then

$$\omega_B := \omega - \pi^*B$$

is symplectic and that, if $B, B' \in \Omega^2(Q)$ are closed and such that $B - B' = dA$, then φ_A (defined in the previous problem) is a symplectomorphism from (T^*Q, ω_B) to $(T^*Q, \omega_{B'})$.

Problem 6: Let $\omega \in \Omega^2(M)$ be a nondegenerate 2-form. For $f \in C^\infty(M)$, let $X_f \in \mathfrak{X}(M)$ be defined by $i_{X_f}\omega = df$. Consider the bracket $\{f, g\} := \omega(X_g, X_f)$. Verify that $d\omega = 0$ if and only if $\{\cdot, \cdot\}$ satisfies the Jacobi identity.

Problem 7: (1) Consider symplectic manifolds (M_i, ω_i) , with Poisson bracket $\{\cdot, \cdot\}_i$, $i = 1, 2$, and let $\phi : M_1 \rightarrow M_2$ be a smooth map.

- (a) Prove that, if ϕ is a diffeomorphism, then it is a Poisson map ($\{\phi^*f, \phi^*g\}_1 = \phi^*(\{f, g\}_2)$ for all $f, g \in C^\infty(M_2)$) if and only if $\phi^*\omega_2 = \omega_1$.
- (b) Find examples of M_1, M_2 and $\phi : M_1 \rightarrow M_2$ such that (1) ϕ is a Poisson map but does not satisfy $\phi^*\omega_2 = \omega_1$; (2) ϕ satisfies $\phi^*\omega_2 = \omega_1$ but is not a Poisson map.

Hint: Consider $\mathbb{R}^2$ and $\mathbb{R}^4$ with their canonical symplectic structures and Poisson brackets, and the maps $\mathbb{R}^2 \rightarrow \mathbb{R}^4$, $(q_1, p_1) \mapsto (q_1, p_1, 0, 0)$, and $\mathbb{R}^4 \rightarrow \mathbb{R}^2$, $(q_1, p_1, q_2, p_2) \mapsto (q_1, p_1)$.

Problem 8:

- (a) Consider $S^2 = \{x \in \mathbb{R}^3 \mid \|x\| = 1\}$ equipped with the area form $\omega_x(u, v) = \langle x, u \times v \rangle$ (where $x \in S^2$, $u, v \in T_x S^2$, and $\times$ is the vector product). Use cylindrical coordinates to prove Darboux's theorem directly in this example.
- (b) More generally: show that on a 2-dimensional manifold, any non-vanishing 1-form can be locally written as fdg , where f and g are smooth functions. Use this fact to give a direct proof of Darboux's theorem in 2 dimensions.

Geometria Simplética 2024, Lista 3

Prof: H. Bursztyn

Entrega 20/09

Problem 1: Consider a symplectic manifold (M^{2n}, ω) with hamiltonian $H \in C^\infty(M)$. Suppose c is a regular value of H . We will show that $M_c = H^{-1}(c)$ inherits a natural volume form, invariant by the hamiltonian flow. We will actually show something more general.

Let $f_1, \dots, f_k \in C^\infty(M)$ be first integrals of the flow of H (i.e., $\{H, f_i\} = 0$). Let $F = (f_1, \dots, f_k) : M \rightarrow \mathbb{R}^k$, and let $c \in \mathbb{R}^k$ be a regular value. Note that $M_c := F^{-1}(c)$ is invariant by the flow of H . We will show that M_c carries a natural invariant volume form.

- a) Take a neighborhood $\mathcal{U}$ of M_c where $df_1, \dots, df_k$ are linearly independent pointwise. Show that the Liouville volume form $(\Lambda_\omega = \omega^n/n!)$ can be written in $\mathcal{U}$ as $\Lambda_\omega = df_1 \wedge \dots \wedge df_k \wedge \sigma$, for some $\sigma \in \Omega^{2n-k}(M)$. We then define a volume form $\Lambda_c := \iota^* \sigma \in \Omega^{2n-k}(M_c)$, where $\iota : M_c \hookrightarrow M$ is the inclusion.

Hint: find σ locally and use partition of unity.

- b) Show that $df_1 \wedge \dots \wedge df_k \wedge \mathcal{L}_{X_H} \sigma = 0$, and use this fact to see that we can write $\mathcal{L}_{X_H} \sigma = \sum_{i=1}^k df_i \wedge \rho_i$. Conclude that Λ_c is invariant by the flow of H .
- c) Show that Λ_c does not depend on the choice σ .

Problem 2: Let M be a symplectic manifold, $\Psi = (\psi^1, \dots, \psi^k) : M \rightarrow \mathbb{R}^k$ a smooth map, and c a regular value. Consider a submanifold $N = \Psi^{-1}(c) \hookrightarrow M$.

- (a) Show that N is coisotropic if and only if $\{\psi^i, \psi^j\}|_N = 0$ for all $i, j = 1, \dots, k$.
- (b) Show that N is symplectic if and only if the matrix (c^{ij}) , with $c^{ij} = \{\psi^i, \psi^j\}$, is invertible for all $x \in N$. In this case, verify that we have the following expression for the Poisson bracket $\{\cdot, \cdot\}_N$ on N (known *Dirac's bracket*):

$$\{f, g\}_N = (\{\tilde{f}, \tilde{g}\} - \sum_{ij} \{\tilde{f}, \psi^i\} c_{ij} \{\psi^j, \tilde{g}\})|_N,$$

where $(c_{ij}) = (c^{ij})^{-1}$, $f, g \in C^\infty(N)$, e $\tilde{f}, \tilde{g} \in C^\infty(M)$ are arbitrary extensions of f, g , respectively. [Hint: we have $TM|_N = TN \oplus TN^\omega$, and projections $q_1 : TM|_N \rightarrow TN$ and $q_2 : TM|_N \rightarrow TN^\omega$; show that $X_f = q_1(X_{\tilde{f}})$, and verify that $q_2(Y) = \sum_{i,j} d\psi^i(Y) c_{ij} X_{\psi^j}$.]

Problem 3: Let $D \subseteq TM$ be a vector subbundle, and let $\text{Ann}(D) \subseteq T^*M$ be its annihilator. Show that D is involutive iff $\text{Ann}(D)$ is a coisotropic submanifold of T^*M .

Problem 4: Consider a smooth map $\phi : Q_1 \rightarrow Q_2$, and let

$$R_\phi := \{((x, \xi), (y, \eta)) \mid y = \phi(x), \xi = (T\phi)^* \eta\} \subset T^*Q_1 \times T^*Q_2.$$

Verify that R_ϕ is a lagrangian submanifold of $T^*Q_1 \times \overline{T^*Q_2}$. Whenever ϕ is a diffeo, what is the relation between R_ϕ and the cotangent lift $\widehat{\phi}$?

Denote by $\Gamma_\phi \subset Q_1 \times Q_2$ the graph of ϕ . What is the relation between $N^*\Gamma_\phi$ (the conormal bundle of Γ_ϕ) and R_ϕ ?

Problem 5: Let M be a manifold and $\omega \in \Omega^k(M)$. Suppose that $\pi : M \rightarrow B$ is a surjective submersion with connected fibers. We say that ω is *basic* (with respect to π) if there exists a form $\bar{\omega} \in \Omega^k(B)$ such that $\pi^*\bar{\omega} = \omega$.

- (a) Show that ω is basic iff $i_X\omega = 0$ and $\mathcal{L}_X\omega = 0$ for all vector fields X tangent to the fibers of π . In particular, if ω is closed, show that it is basic if $\ker(T\pi) \subseteq \ker(\omega)$ (pointwise in M).
- (b) Suppose that ω is a closed 2-form on M and $\ker(T\pi) = \ker(\omega)$. Show that $\omega = \pi^*\bar{\omega}$ and $\bar{\omega} \in \Omega^2(B)$ is symplectic.
- (c) (Application to reduction) Let (M, ω) be a symplectic manifold and $\iota : N \hookrightarrow M$ a submanifold such that $D = TN \cap TN^\omega \subset TN$ has constant rank (e.g., N could be coisotropic). We saw in class that D is an integrable distribution (by Frobenius); suppose that the leafspace $B := N/\sim$ is smooth so that the natural projection $\pi : N \rightarrow B$ is a submersion. Show that B inherits a unique symplectic form ω_{red} with the property that $\pi^*\omega_{red} = \iota^*\omega$.

Geometria Simplética 2024, Lista 4

Prof. H. Bursztyn

Entrega dia 30/09

Problem 1: Let M be a compact, connected, orientable n -dimensional manifold. Let $\Lambda_0, \Lambda_1 \in \Omega^n(M)$ be two volume forms on M such that $\int_M \Lambda_0 = \int_M \Lambda_1$. Show that there is a diffeomorphism $\phi \in \text{Dif}(M)$ such that $\phi^*(\Lambda_1) = \Lambda_0$.

Hint: Show that $\Lambda_t = (1 - t)\Lambda_0 + t\Lambda_1$ is a volume form $\forall t \in [0, 1]$, and note that the equality of total volumes implies that Λ_0 and Λ_1 define the same cohomology class (justify this fact). Then apply Moser's method.

Problem 2: Give an example of two symplectic forms on $\mathbb{R}^4$ that induce the same orientation, but admit a convex combination that is degenerate. Is it possible to find an example like that, but admitting another path of *symplectic* forms from one to the other? What happens if we consider $\mathbb{R}^2$ instead of $\mathbb{R}^4$?

Problem 3: This problem is useful for Problem 4. (Cf. Problem 7 in lista 1)

Let (V, Ω) be a symplectic vector space (or, more generally, a symplectic vector bundle), and let $W \subseteq V$ be a coisotropic subspace (resp., coisotropic subbundle).

- a) Let E be a complement of W^Ω in W (i.e., $W = W^\Omega \oplus E$). Show that the restriction of Ω to E is nondegenerate.
- b) Let J be a Ω -compatible complex structure, with g the associated inner product. Show that Ω induces an identification of $J(W^\Omega) = W^\perp$ with $(W^\Omega)^*$. Taking E as the orthogonal complement (with respect to g) to W^Ω in W , show that the identification

$$V \cong E \oplus (W^\Omega \oplus (W^\Omega)^*), \quad (1)$$

is an isomorphism of symplectic vector spaces (resp. bundles) – on the right-hand side, E is equipped with its induced symplectic form (see (a) above) and $W^\Omega \oplus (W^\Omega)^*$ with its canonical symplectic form.

Problem 4: Prove the following generalization of Weinstein's lagrangian neighborhood theorem to coisotropic submanifolds (due to Gotay, 1982): Let (M_0, ω_0) and (M_1, ω_1) be symplectic manifolds, and $\iota_0 : Q \rightarrow M_0$, $\iota_1 : Q \rightarrow M_1$ be coisotropic embeddings. If $\iota_0^* \omega_0 = \iota_1^* \omega_1$, then there exist open neighborhoods $\mathcal{U}_0$ and $\mathcal{U}_1$ of Q , in M_0 and M_1 , and a diffeomorphism $\varphi : \mathcal{U}_0 \rightarrow \mathcal{U}_1$ such that $\varphi(p) = p$ for all $p \in Q$ and $\varphi^* \omega_1 = \omega_0$.

Hint: For $i = 0, 1$, use compatible almost complex structures to write $TM_i|_Q$ as in (??); conclude that there is an isomorphism $(TM_0|_Q, \omega_0) \rightarrow (TM_1|_Q, \omega_1)$ of symplectic vector bundles, so you can apply Weinstein's generalized Darboux theorem.

Geometria Simplética 2024, Lista 5

Prof. H. Bursztyn

Entrega dia 11/10

Problem 1: Let G be a Lie group. Let $X : G \rightarrow TG$ be a section of the projection $TG \rightarrow G$, not necessarily smooth. Show that if X is left invariant (i.e., $dL_g(X) = X \circ L_g$ for all $g \in G$), then X is automatically smooth.

Conclude that an analogous result holds for differential forms: if a section $\eta : G \rightarrow \wedge^k T^*G$ is left invariant ($L_g^* \eta = \eta$), then η is a smooth k -form. Check that an analogous result holds for G -invariant forms on a homogeneous manifold.

Problem 2: (a) Prove that any connected Lie group G is generated (as a group) by any open neighborhood U of the identity element (i.e., $G = \cup_{n=1}^{\infty} U^n$). (b) Suppose that two Lie group homomorphisms $\varphi, \psi : G \rightarrow H$ are such that $d\varphi|_e = d\psi|_e$. Show that φ and ψ coincide on the connected component of G containing the identity e .

Problem 3: Consider the Lie groups $SU(2) = \{A \in M_2(\mathbb{C}) \mid AA^* = \text{Id}, \det(A) = 1\}$ and $SO(3) = \{A \in M_3(\mathbb{R}) \mid AA^t = \text{Id}, \det(A) = 1\}$.

a) Show that

$$SU(2) = \left\{ \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix}, \quad a, b \in \mathbb{C}, |a|^2 + |b|^2 = 1 \right\}.$$

Conclude that, as a manifold, $SU(2)$ is diffeomorphic to S^3 (hence it is simply connected).

Recall the definition of the quaternions $\mathbb{H}$. Show that the sphere S^3 , seen as quaternions of norm 1, inherits a Lie group structure, with respect to which it is isomorphic to $SU(2)$.

b) Verify that

$$\mathfrak{su}(2) = \left\{ \begin{pmatrix} i\alpha & \beta \\ -\bar{\beta} & -i\alpha \end{pmatrix}, \quad \alpha \in \mathbb{R}, \beta \in \mathbb{C} \right\}.$$

Consider the identification $\mathfrak{su}(2) \cong \mathbb{R}^3$, that takes the element in $\mathfrak{su}(2)$ determined by α, β to the vector $(\alpha, \text{Re}\beta, \text{Im}\beta)$ in $\mathbb{R}^3$. Observe that, with respect to this identification, $\det$ in $\mathfrak{su}(2)$ corresponds to $\|\cdot\|^2$ in $\mathbb{R}^3$.

c) Verify that each element $A \in SU(2)$ defines a linear transformation on the vector space $\mathfrak{su}(2)$ by conjugation: $B \mapsto ABA^{-1}$. Show that, with the identification $\mathfrak{su}(2) \cong \mathbb{R}^3$, we obtain a representation (i.e., a linear action) of $SU(2)$ on $\mathbb{R}^3$ that is norm preserving. Conclude that we have a homomorphism $\phi : SU(2) \rightarrow O(3)$, verifying that its image is $SO(3)$ and its kernel is $\{\text{Id}, -\text{Id}\}$.

d) Conclude that $SU(2) \cong S^3$ is a double cover of $SO(3)$ (hence it is its universal cover, since it's simply connected), and the covering map identifies antipodal points of S^3 . Hence, as manifolds, $SO(3)$ is identified with $\mathbb{R}P^3$.

Problem 4: Let $\mathfrak{g}$ be the Lie algebra of a Lie group G , and let $k : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{R}$ be a symmetric bilinear form that is Ad-invariant (i.e., $k(\text{Ad}_g(u), \text{Ad}_g(v)) = k(u, v)$ for $g \in G$).

a) Show that the map $k^\sharp : \mathfrak{g} \rightarrow \mathfrak{g}^*$, $k^\sharp(u)(v) = k(u, v)$, is G -equivariant:

$$k^\sharp \circ \text{Ad}_g = (\text{Ad}^*)_g \circ k^\sharp, \quad \forall g \in G. \quad (1)$$

[recall: $(\text{Ad}^*)_g := (\text{Ad}_{g^{-1}})^*$]. In particular, when k is nondegenerate (i.e., $k^\sharp$ is an isomorphism), the adjoint and coadjoint actions are equivalent.

b) Verify that (1) implies that $k([w, u], v) = -k(u, [w, v])$, $\forall u, v, w \in \mathfrak{g}$, and that both conditions are equivalent when G is connected.

Problem 5: For a Lie algebra $\mathfrak{g}$, there is always a canonical bilinear form $k : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{R}$, called *Killing form*, given by:

$$k(u, v) = \text{tr}(\text{ad}_u \text{ad}_v).$$

(recall: $\text{ad}_u : \mathfrak{g} \rightarrow \mathfrak{g}$, $\text{ad}_u(v) = [u, v]$.)

a) Note that k is symmetric, and check that it is Ad-invariant.

b) A Lie algebra is called *semi-simple* if k is nondegenerate. Show that $\mathfrak{so}(3)$ is semi-simple.

Problem 6: Consider the linear isomorphism $\mathbb{R}^3 \rightarrow \mathfrak{so}(3)$, given by

$$v = (x, y, z) \mapsto \widehat{v} := \begin{pmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0 \end{pmatrix}$$

a) Describe the Lie bracket on $\mathbb{R}^3$ induced by the commutator in $\mathfrak{so}(3)$, and the inner product in $\mathfrak{so}(3)$ that corresponds to the canonical inner product in $\mathbb{R}^3$.

b) Describe the $SO(3)$ -action on $\mathbb{R}^3$ corresponding to the adjoint action, its orbits, as well as its infinitesimal generators. Find (without any calculation!) a description of the coadjoint action on $\mathbb{R}^3$ (identified with $(\mathbb{R}^3)^*$ through the canonical inner product).

Problem 7: Let (V, Ω) be a symplectic vector space, and consider $H := V \times \mathbb{R} = \{(v, t)\}$. This space H , with the multiplication

$$(v_1, t_1) \cdot (v_2, t_2) = (v_1 + v_2, \frac{1}{2}\Omega(v_1, v_2) + t_1 + t_2),$$

is a Lie group, called the *Heisenberg group* (find the identity elements and inverses in H).

(a) Show (directly from the conjugation formula in H) that $\text{Ad}_{(v,t)}(X, r) = (X, r + \Omega(v, X))$, for $(X, r) \in \mathfrak{h} = \text{Lie}(H) = V \times \mathbb{R}$. Describe the adjoint orbits, verifying that their possible dimensions are zero and one.

(b) Verify that $\text{ad}_{(Y,s)}(X, r) = (0, \Omega(Y, X))$. [Recalling that $\text{ad}_{(Y,s)}(X, r) = [(Y, s), (X, r)]$, we obtain a formula for the Lie bracket in $\mathfrak{h}$.]

(c) Describe the coadjoint action of H on $\mathfrak{h}^* = V^* \times \mathbb{R}^*$ and its orbits, analyzing the possible dimensions.

Geometria Simplética 2024, Lista 6

Prof. H. Bursztyn

Entrega dia 23/10

Problem 1:

- (a) Consider hamiltonian actions of G on two symplectic manifolds (M_i, ω_i) , $i = 1, 2$, with moment maps $\mu_i : M_i \rightarrow \mathfrak{g}^*$, $i = 1, 2$. Show that the diagonal action of G on $M_1 \times M_2$ ($g(x_1, x_2) \mapsto (gx_1, gx_2)$) is hamiltonian, with moment map $\mu : M_1 \times M_2 \rightarrow \mathfrak{g}^*$, $\mu(x_1, x_2) = \mu_1(x_1) + \mu_2(x_2)$.
- (b) Suppose that $G \curvearrowright M$ is a hamiltonian action with moment map μ , and let $H \subseteq G$ be a Lie subgroup. Show that the restriction of the action to H , $H \curvearrowright M$, is hamiltonian with moment map $\iota^* \circ \mu$, where $\iota : \mathfrak{h} \rightarrow \mathfrak{g}$ is the inclusion.

Problem 2: Consider the group $SO(3)$ acting on $T^*\mathbb{R}^3$ by the cotangent lift of the usual action of $SO(3)$ on $\mathbb{R}^3$.

- a) For $u \in \mathfrak{so}(3)$, compute the corresponding infinitesimal generator $u_{T^*\mathbb{R}^3} \in \mathfrak{X}(T^*\mathbb{R}^3)$.
- b) Identify $\mathfrak{so}(3)$ with $\mathbb{R}^3$ (as in Lista 6). Show that, with this identification, we have $u_{T^*\mathbb{R}^3}(q, p) = (u \times q, u \times p)$.
- c) Identifying $\mathfrak{so}(3)^* \cong (\mathbb{R}^3)^* \cong \mathbb{R}^3$ using the usual inner product, show that the moment map for the action of $SO(3)$ on $T^*\mathbb{R}^3$ is $\mu(q, p) = q \times p$. Conclude (by Noether's theorem) that if $V \in C^\infty(\mathbb{R}^3)$ is $SO(3)$ -invariant, then the flow of the hamiltonian $H(q, p) = \frac{p^2}{2m} + V(q)$ preserves “angular momentum” $q \times p$.

Problem 3: Consider $G = \mathbb{R}^2$ acting on $\mathbb{R}^2$ by $g \cdot (x, y) = (x + a, y + b)$, where $g = (a, b)$. Show that this action is *weakly* hamiltonian (i.e., there exists $\mu : M \rightarrow \mathfrak{g}^*$ such that $i_{u_M}\omega = d\langle \mu, u \rangle$) but it does *not* admit an equivariant moment map.

Problem 4: Consider a weakly hamiltonian G -action on (M, ω) , with moment map $\mu : M \rightarrow \mathfrak{g}^*$ (i.e., not necessarily equivariant). We now see two independent cases in which we can always find a moment map which is equivariant.

- a) For each $g \in G$, define $g \cdot \mu := (\text{Ad}^*)_g(\mu \circ g^{-1})$ (in such a way that μ is equivariant if and only if $g \cdot \mu = \mu$ for all g). Show that $g \cdot \mu$ is also a moment map (not necessarily equivariant) for the action.
- b) Suppose that G is compact. In this case, we can take a left-invariant volume form Λ on G (i.e., $L_g^*\Lambda = \Lambda$) satisfying $\int_G \Lambda = 1$ (why?). Consider the “average” $\bar{\mu} := \int_G g \cdot \mu$ (integral with respect to Λ). Show that $\bar{\mu}$ is an equivariant moment map.
- c) Suppose that M is compact and connected. Then there is an equivariant moment map.

(Hint: Note that we can take μ normalized so that $\int_M \mu = 0$ (integral with respect to the Liouville volume Λ_ω). Verify, using that M is connected, that this normalization uniquely characterizes the moment map. Conclude that μ is equivariant by showing that $\int_M g \cdot \mu = 0$, for all $g \in G$.)

Problem 5: Consider the torus $\mathbb{T}^n$ acting on $\mathbb{C}^n$ (the canonical symplectic form reads $\frac{i}{2} \sum_j dz_j \wedge d\bar{z}_j$) by:

$$(e^{i\theta_1}, \dots, e^{i\theta_n}) \cdot (z_1, \dots, z_n) = (e^{ik_1\theta_1} z_1, \dots, e^{ik_n\theta_n} z_n),$$

where $k_1, \dots, k_n \in \mathbb{Z}$ are fixed.

- a) Show that this action is hamiltonian, with moment map $\mu : \mathbb{C}^n \rightarrow (\mathfrak{t}^n)^* \cong \mathbb{R}^n$,

$$\mu(z_1, \dots, z_n) = -\frac{1}{2}(k_1|z_1|^2, \dots, k_n|z_n|^2).$$

- b) Conclude (see Problem 1) that the action of S^1 on $\mathbb{C}^n$ given by multiplication by $e^{i\theta}$ on each coordinate is hamiltonian with moment map $\mu : \mathbb{C}^n \rightarrow \mathbb{R}$, $\mu(z) = -\frac{1}{2}|z|^2$.

Problem 6: Consider the usual action of $U(n)$ on $\mathbb{C}^n$.

- a) Writing elements $U \in U(n)$ in the form $A + iB$, check that the action of U on $\mathbb{R}^{2n}$ ($\cong \mathbb{C}^n$) is given by the linear symplectomorphism

$$\begin{pmatrix} A & -B \\ B & A \end{pmatrix}.$$

The Lie algebra $\mathfrak{u}(n)$ consists of anti-hermitian matrices $u = \xi + i\eta$, with $\xi = -\xi^t \in M_n(\mathbb{R})$, $\eta = \eta^t \in M_n(\mathbb{R})$. Show that the infinitesimal generator of $u \in \mathfrak{u}(n)$ is hamiltonian with respect to

$$\mu^u(z) = -\frac{1}{2}\langle x, \eta x \rangle + \langle y, \xi x \rangle - \frac{1}{2}\langle y, \eta y \rangle,$$

where $z = x + iy$, $x, y \in \mathbb{R}^n$, and $\langle \cdot, \cdot \rangle$ is the usual inner product.

- b) Show that $\mu^u(z) = \frac{1}{2}iz^*uz = \frac{1}{2}i\text{tr}(zz^*u)$.
c) Identify $\mathfrak{u}(n)$ with $\mathfrak{u}(n)^*$ through the inner product $(A, B) = \text{tr}(A^*B)$. Let $\mu : \mathbb{C}^n \rightarrow \mathfrak{u}(n)$,

$$\mu(z) = \frac{i}{2}zz^*.$$

Here $z \in \mathbb{C}^n$ is viewed as a $n \times 1$ matrix. Show that μ is equivariant (recall what the adjoint and coadjoint actions are) and conclude that μ is a moment map for the $U(n)$ -action.

- d) Consider the action of $U(k)$ on the space $\mathbb{C}^{k \times n}$ (with the canonical symplectic form), viewed as $k \times n$ matrices. Identify $\mathfrak{u}(k)$ with its dual as in item c). Show that a moment map for this action is

$$\mu(A) = \frac{i}{2}AA^* - \frac{i\text{Id}}{2}.$$

(Hint: combine problem 1(a) and the previous items, the constant factor is just for convenience.)

Verify that $\mu^{-1}(0)/U(k)$ is naturally identified with the Grassmannian of k -planes in $\mathbb{C}^n$ (which hence acquires a symplectic form from symplectic reduction).

Problem 7: Consider a hamiltonian action $\psi : G \curvearrowright (M, \omega)$, with moment map $\mu : M \rightarrow \mathfrak{g}^*$. Consider the co-moment map $\hat{\mu} : \mathfrak{g} \rightarrow C^\infty(M)$, $\hat{\mu}(u) = \langle \mu, u \rangle$, and consider $C^\infty(M)$ equipped with the Poisson bracket.

- (a) We saw in class that the equivariance of μ implies that $\hat{\mu}$ is an anti-homomorphism of Lie algebras. Show that the converse holds when G is connected.
- (b) Show that for G connected, the fact that $\psi_g^* \omega = \omega$ follows from the condition $i_{u_M} \omega = d\langle \mu, u \rangle$.

Problem 8: Let G be a Lie group and consider the action by multiplication on the left: $G \times G \rightarrow G$, $(g, a) \mapsto L_g(a) = g.a$. Take the G -action on T^*G by cotangent lift, $\psi : G \times T^*G \rightarrow T^*G$.

- (a) Consider the action of G_ξ on G by left multiplication, and the map $q : G \rightarrow \mathcal{O}_\xi$, $g \mapsto \text{Ad}_{g^{-1}}^* \xi$ (where $\mathcal{O}_\xi$ is the coadjoint orbit thru ξ). Note that we have an induced bijection $G/G_\xi \rightarrow \mathcal{O}_\xi$. (It is a general fact that the quotient G/G_ξ is naturally a smooth manifold, and we equip $\mathcal{O}_\xi$ with the smooth structure for which this bijection is a diffeomorphism.)

Verify that $q(R_h(g)) = \text{Ad}_{h^{-1}}^*(q(g))$, and conclude that $dq(u^L|_g) = u_{\mathfrak{g}^*}|_{q(g)}$ and $dq(u^R|_g) = (\text{Ad}_{g^{-1}}(u))_{\mathfrak{g}^*}|_{q(g)}$.

- (b) Verify that $\mu : T^*G \rightarrow \mathfrak{g}^*$, $\mu(\zeta_g) = (d_e R_g)^* \zeta_g$, is a moment map for ψ . Note that any $\xi \in \mathfrak{g}^*$ is a regular value for μ and that the natural projection $\pi : \mu^{-1}(\xi) \rightarrow G$ is a diffeomorphism. Show also that the projection induces a diffeomorphism:

$$(\mu^{-1}(\xi)/G_\xi) \xrightarrow{\sim} (G/G_\xi).$$

Using the identification $G/G_\xi \cong \mathcal{O}_\xi$ of (a), find an expression for the resulting diffeomorphism $\varphi : (\mu^{-1}(\xi)/G_\xi) \xrightarrow{\sim} \mathcal{O}_\xi$.

- (c) Show that $-\varphi^* \omega_{kks} = \omega_{red}$, that is, the reduced space $(\mu^{-1}(\xi)/G_\xi, \omega_{red})$ is symplectomorphic to $(\mathcal{O}_\xi, -\omega_{kks})$.

*Hints: recall that, for the projection $\pi_\xi : \mu^{-1}(\xi) \rightarrow \mathcal{O}_\xi$ and inclusion $i_\xi : \mu^{-1}(\xi) \rightarrow T^*G$, we must show that $\pi_\xi^* \omega_{kks} = i_\xi^* d\alpha_{\text{tau}}$; note that it suffices to check this equality on vectors X tangent to $\mu^{-1}(\xi)$ satisfying $\pi_* X = u^R$, for $u \in \mathfrak{g}$ (why?).*

Problem 9: (“Shift trick”) Let (M, ω, μ) be a hamiltonian G -space. Take a coadjoint orbit $\overline{\mathcal{O}_\xi}$ (with symplectic form $-\omega_{kks}$). Verify that the diagonal G -action on $M \times \overline{\mathcal{O}_\xi}$ is hamiltonian, with moment map

$$\hat{\mu} : M \times \overline{\mathcal{O}_\xi} \rightarrow \mathfrak{g}^*, \quad \hat{\mu}(x, \eta) = \mu(x) - \eta,$$

and that ξ is a regular value for μ if and only if 0 is a regular value for $\hat{\mu}$.

Note that we have a natural inclusion $j : \mu^{-1}(\xi) \hookrightarrow \hat{\mu}^{-1}(0)$, $x \mapsto (x, \xi)$. Show that this inclusion induces a diffeomorphism $\mu^{-1}(\xi)/G_\xi \xrightarrow{\sim} \hat{\mu}^{-1}(0)/G$ preserving the reduced symplectic forms.

Symplectic Geometry 02/2024, Homework 7

Professor: Leonardo Macarini

Deadline: 22/11

Problem 1:

- (a) Let $\alpha \in \Omega^1(N)$ be a contact form. Consider $N \times \mathbb{R}$ equipped with the 2-form $d(e^t\alpha)$ (where t is the coordinate on $\mathbb{R}$). Verify that this 2-form is symplectic, and conclude that any (N, α) can be viewed as a hypersurface of contact type of a symplectic manifold.
- (b) On the other hand: Let (M, ω) be symplectic and $\iota : S \hookrightarrow M$ a hypersurface of contact type, with contact form α . Suppose that S is compact. Show that there is a neighborhood U of S in M that is symplectomorphic to a neighborhood of S in its symplectification, $(S \times (-\epsilon, \epsilon), d(e^t\alpha))$, for an $\epsilon > 0$.

Hints:

- Consider the conformally symplectic vector field X in a neighborhood of S (such that $\alpha = \iota^*(i_X\omega)$); use the tubular neighborhood theorem to obtain an identification $\psi : U \xrightarrow{\sim} S \times (-\epsilon, \epsilon)$, in such a way that X corresponds to $\frac{\partial}{\partial t}$.
 - Let Y be the Reeb vector field on S . For each $x \in S$, write $T_x M = W_1 \oplus W_2$, where $W_1 = \ker(\alpha) \subset T_x M$ and $W_2 \subset T_x M$ is the subspace generated by $X(x)$ and $Y(x)$. Verify that W_1 and W_2 are symplectically orthogonal with respect to ω , and also with respect to $\psi^*d(e^t\alpha)$.
 - Use the decomposition $T_x M = W_1 \oplus W_2$ to verify that ω and $\psi^*d(e^t\alpha)$ coincide on $TM|_S$. Use the Darboux-Weinstein theorem.
- (c) Let $\xi \in \Omega^1(N)$ be a contact form on N and $D = \ker(\xi) \subset TN$. Let $L \subset N$ be a submanifold such that $TL \subset D|_L$. (1) Check that $T_x L$ is an isotropic subspace of the symplectic vector space $(D_x, d\xi|_x)$ for all $x \in L$, so $\dim(L) \leq \frac{1}{2}\text{rank}(D)$. In case of equality, we call L **legendrian**. (2) Verify that L is legendrian iff $L \times \mathbb{R}$ is a lagrangian submanifold of the symplectization $N \times \mathbb{R}$.
- (d) We saw that S^{2n-1} has contact structure $\alpha = \iota^*(\frac{1}{2} \sum_{i=1}^n q^i dp_i - p_i dq^i)$, where $\iota : S^{2n-1} \hookrightarrow \mathbb{R}^{2n}$ is the inclusion. Show that its symplectization is symplectomorphic to $\mathbb{R}^{2n} - \{0\}$ (with the canonical symplectic form).

Problem 2: Show the Darboux theorem for contact manifolds: Given contact manifold (N^{2n-1}, α) , around any point there exist local coordinates $q^1, \dots, q^{n-1}, p_1, \dots, p_{n-1}, z$, such that $\alpha = \sum_i q^i dp_i + dz$.

Hint: One can, for example, use symplectization and adapt the symplectic Darboux theorem

Problem 3: Let N be a closed symplectic manifold (e.g. S^2), and let $M = N \times \mathbb{R}^2$, equipped with the product symplectic structure. Show that the hypersurface $C = N \times \{z \in \mathbb{R}^2 \mid \|z\| = 1\} \hookrightarrow M$ is not of contact type.

Problem 4: The *manifold of contact elements* of an n -dimensional manifold X is $\mathcal{C} = \{(x, \chi_x); x \in X \text{ and } \chi_x \text{ is a hyperplane in } T_x X\}$. On the other hand, the projectivization of the cotangent bundle of X is $\mathbb{P}^*X = (T^*X \setminus \text{zero section}) / \simeq$, where $(x, \xi) \simeq (x, \xi')$ whenever $\xi = \lambda \xi'$ for some $\lambda \in \mathbb{R} \setminus \{0\}$.

- (a) Show that $\mathcal{C}$ is naturally isomorphic to $\mathbb{P}^*X$ as a bundle over X .
- (b) There is on $\mathcal{C}$ a canonical field of hyperplanes $\mathcal{H}$: $\mathcal{H}$ at the point $p = (x, \chi_x) \in \mathcal{C}$ is the hyperplane $\mathcal{H}_p = (d\pi_p)^{-1}\chi_x$, where $\pi : \mathcal{C} \rightarrow X$ is the projection. Therefore, by item (a), $\mathcal{H}$ induces a field of hyperplanes $\mathbb{H}$ on $\mathbb{P}^*X$. Describe $\mathbb{H}$.
- (c) Check that $(\mathbb{P}^*X, \mathbb{H})$ is a contact manifold, and therefore $(\mathcal{C}, \mathcal{H})$ is a contact manifold.
- (d) What is the symplectization of $\mathcal{C}$?

Problem 5: Let (M, α) be a contact manifold with contact structure $\xi = \ker \alpha$. A *contact vector field* X on M is a vector field whose (linearized) flow preserves ξ .

- (a) Let R_α be the Reeb vector field of α . Prove that R_α is a contact vector field.
- (b) Suppose that X is a contact vector field transverse to ξ . Show that it can be written as a Reeb vector field for some 1-form α_X defining the contact structure ξ .

Problem 6: Consider, in $\mathbb{R}^{2n}$, the ellipsoid

$$S_{(r_1, \dots, r_n)} = \{(q_1, \dots, q_n, p_1, \dots, p_n) \in \mathbb{R}^{2n} \mid \sum_{j=1}^n \frac{q_j^2 + p_j^2}{r_j^2} = 1\},$$

where $r_1, \dots, r_n$ are positive. We saw that S is a hypersurface of contact type. Compute its Reeb vector field and describe its flow. Describe general conditions on $r_1, \dots, r_n$ so that: (1) there are only n periodic orbits; (2) there are infinitely many periodic orbits.

Problem 7: Consider the unit sphere $S^3 \subset \mathbb{R}^4$ endowed with the contact form induced by the Liouville form. Show that its Reeb flow has a global surface of section given by a disk, that is, there is an embedding $\psi : D^2 \rightarrow S^3$ such that $\psi(\partial D^2)$ is a closed Reeb orbit, the Reeb vector field is transverse to $\psi(\text{Int } D^2)$ and every Reeb orbit not contained in $\psi(\partial D^2)$ intersects $\psi(\text{Int } D^2)$ infinitely many times in the future and the past.